

Set up with Anaconda

For students who already use Anaconda or Miniconda on Windows, macOS or Linux.

What you are setting up: an *environment* is a separate Python installation with its own packages. Conda creates and activates the course environment, named `esbmtk-practicals`. JupyterLab is the interface you open in your browser; a *kernel* is the Python process that runs notebook cells. Your R installation stays unchanged. You do not need `uv` or another Python installer.

1 Download the supplied folder and open a terminal

Download the folder supplied by your instructor and extract the ZIP. It may be the setup-only package or, later, the complete course repository. On Windows, right-click it and choose **Extract All**. On macOS, double-click it. Work in the extracted folder, not inside the ZIP.

The folder must contain **environment-anaconda.yml**, **scripts**, **setup_assets** and **JUPYTER_BASICS.ipynb**. The YAML file is a package recipe supplied by your instructor; you do not need to write it.

Windows: click Start or the Windows search bar, type **Anaconda Prompt**, and open it. This terminal is already prepared to use Conda. A terminal is simply a window where you type commands. You do not need to open Navigator first.

macOS: press **Command+Space**, type **Terminal**, and press Return. **Linux:** open Terminal (often Ctrl+Alt+T). Use a terminal already configured for your Anaconda or Miniconda installation.

Type this command and press Enter:

```
conda --version
```

A version number confirms Conda is available. If it is not recognised, use Anaconda Prompt on Windows. On macOS/Linux, ask your instructor to help connect your existing Conda installation to the terminal.

2 Know where to type commands

Run terminal commands **one line at a time**, pressing Enter and waiting for each to finish. They are not R code and do not go in RStudio or a notebook cell. Do not copy the prompt before a command, such as `(base) C:\Users\Alex>`.

The first installation needs internet access. **Test status:** a fresh environment created from the setup-only ZIP passed the package, numerical, chemistry, workbook, plotting, kernel, notebook and JupyterLab checks on Windows/Python 3.14. macOS/Linux still require instructor pilots. Reference: [Conda environments](#).

Install and open the course

Keep using the Anaconda Prompt or Terminal window from page 1.

3 Tell the terminal which folder to use

The terminal's **current folder** can differ from the folder shown in your file browser. `cd` means change directory (folder). If Alex extracted the course into Downloads, the command would be:

Windows Anaconda Prompt: `/d` also changes the drive if needed.

```
cd /d "C:\Users\Alex\Downloads\ESBMTK-practicals"
```

macOS / Linux:

```
cd "$HOME/Downloads/ESBMTK-practicals"
```

Use your actual folder name and location, including a suffix such as `-main`. On Windows, open the folder in File Explorer, click its address bar and copy that path between the quotes after `cd /d`.

Check the folder: on Windows, run `cd` by itself to print the location, then `dir` to list files. On macOS/Linux, use `pwd`, then `ls`. Find **environment-anaconda.yml**, **scripts**, **setup_assets** and **JUPYTER_BASICS.ipynb** before continuing. "From the supplied folder" means this terminal location.

4 Create, activate and check the environment

```
conda env create -f environment-anaconda.yml
conda activate esbmtk-practicals
python -m pip check
python scripts/check_environment.py
```

Creation downloads Python and the course packages; wait for it to finish. Activation selects them for this terminal; the prompt should show **(esbmtk-practicals)**. If this course environment already exists, skip creation and run the other three commands.

Expect **No broken requirements found**, then **Environment check passed**. **Next: start JupyterLab**. The second check tests numerical calculations, chemistry, access to the supplied workbook probe and plotting. Stop and seek help if either check fails.

5 Register the course kernel and start JupyterLab

Stay in this activated terminal and course folder. Register the kernel once:

```
python -m ipykernel install --sys-prefix --name esbmtk-practicals
python -m jupyterlab
```

JupyterLab opens in your browser. Keep the terminal open; its prompt will not return while the server runs. If no browser opens, copy the local `http://localhost:.../lab` or `http://127.0.0.1:.../lab` link printed there into your browser. Keep the full link private.

Check success and start working

The steps below take place in JupyterLab, then in the terminal when you finish.

6 Complete the JupyterLab setup check

In JupyterLab's left file browser, double-click **JUPYTER_BASIC.ipynb**. Choose **esbmtk-practicals** (or **ESBMTK practicals** if registered with an earlier guide) if JupyterLab asks for a kernel. Follow the notebook from top to bottom, or select **Run > Run All Cells**. The last cell should print:

```
Jupyter setup check passed
```

The notebook also introduces the cell shortcuts used during the practicals. **If both terminal checks and this notebook check pass, setup is complete.** You do not repeat creation or kernel registration each time.

Open or create notebooks while JupyterLab is running

Use the left file browser to double-click any notebook supplied by your instructor. If it asks for the instructor's ESBMTK314 kernel, choose your course kernel instead. Create another notebook via **File > New > Notebook**. No terminal command is needed for each notebook. Save with Ctrl+S (Cmd+S on macOS).

When you finish: save and shut down

Save your notebooks. Return to the terminal running JupyterLab, press **Ctrl+C**, and, if asked to shut down, type **y** and press Enter. Wait for the normal prompt to return, then close the browser tab and terminal. This stops notebook kernels: variables in memory are lost, but saved notebooks remain on disk.

Closing the browser tab alone does not normally stop JupyterLab. Reopen the local link still shown in its terminal if you only closed the tab; you do not need another server.

Next time: open JupyterLab again

Open Anaconda Prompt or your Conda-enabled Terminal, repeat `cd` to the supplied folder (page 2), then run:

```
conda activate esbmtk-practicals
python -m jupyterlab
```

Do this only when the server has stopped, including after restarting your laptop. Open your saved notebooks and rerun the cells needed to recreate variables.

If a step fails

File not found: check the terminal folder (page 2). **Wrong Python or missing package:** activate the course environment, launch Jupyter there and select its kernel. **Windows Python crash:** use the activated prompt. For other errors, send your instructor the full error text, command and operating system.